Algebra 1
Unit 11
Lesson 1 Practice

Name___Answer Key______
Date______________________
Period______________________

1. In the equation $y = a(b)^x$, a represents the
- ○ growth factor
- ● initial amount

and b represents the
- ● growth factor.
- ○ initial amount.

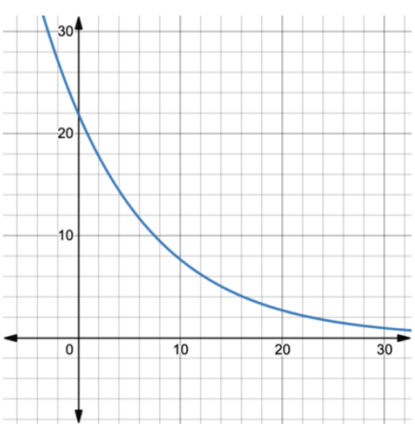
2. When b is between than 0 and 1, the equation models
- ○ exponential growth.
- ● exponential decay.

3. When b is greater than 1, the equation models
- ● exponential growth.
- ○ exponential decay.

For questions 4-8, identify if each representation of a function is an example of exponential growth or exponential decay. Then justify your selection.

4. $y = 142(1.05)^x$ represents ___exponential growth___ because…

$b > 1$ and percent rate of change is greater than 100%.

5. 

represents ___exponential decay___ because… y-values decrease while x-values increase.

6.

x	y
-1	$\frac{1}{5}$
0	$\frac{1}{2}$
1	$\frac{5}{4}$
2	$\frac{25}{8}$

$\times 2.5$

$\times 2.5$

$\times 2.5$

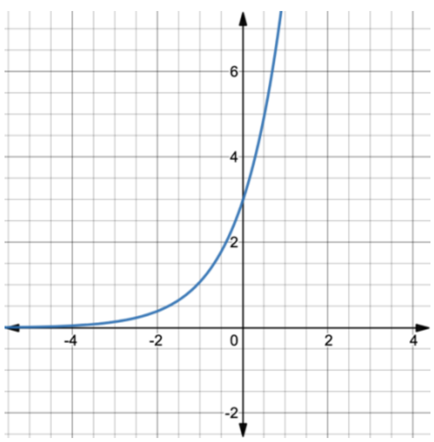
represents ___exponential growth___ because…

increasing at a rate 2.5 times the previous value (a factor of 2.5)

7. $y = 3(1 - 0.14)^x$ represents ___exponential decay___ because…

$3(.86)^x$ the value of b is < 1.

8.



represents ___exponential growth___ because…

The graph increases at a rapid rate. As x increases, y increases at a rapid rate.

9. Write an equation in $y = a(b)^x$ form that represents exponential growth.

$y = 2(1.25)^x$

$b > 1$

10. Write an equation in $y = a(b)^x$ form that represents exponential decay.

$y = 2(.94)^x$

$0 < b < 1$